LESSON 5.10

Volume of Rectangular Prisms

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the volume of a rectangular prism, including ones with fractional edge lengths, using base area × height.

WHY THIS MATTERS Your class is building a time capsule to bury on school grounds. The capsule is a rectangular prism that is 2 feet long, 1.5 feet wide, and 1 foot tall. You need to figure out how much stuff can fit inside — that means calculating the volume!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Volume *Spanish: Volumen*	How much space is inside a solid shape.	A box 2 × 1.5 × 1 = 3 ft³ — it holds 3 cubic feet of stuff
Rectangular prism *Spanish: Prisma rectangular*	A solid box shape with six flat rectangle sides.	A tissue box, a fish tank, a brick — all rectangular prisms
Cubic units *Spanish: Unidades cúbicas*	The units used to measure space inside, like cubic inches.	1 ft³ = a cube that is 1 foot on every edge
Dimensions *Spanish: Dimensiones*	How long, how wide, and how tall a shape is.	A box with dimensions 4 × 3 × 2 means l = 4, w = 3, h = 2
Base area *Spanish: Área de la base*	The area of the bottom of a solid. Volume = base area × height.	If the base is 6 × 4 = 24 cm² and h = 3, then V = 24 × 3 = 72 cm³

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

My capsule is 2 ft long, 1.5 ft wide, and 1 ft tall. I write V = length × width × height.

1. I put in the numbers: V = 2 × 1.5 × 1.
2. First I multiply 2 × 1.5 = 3.
3. Then 3 × 1 = 3.
4. So the volume is 3 ft³. A fractional edge like 1.5 did not change my steps at all.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now try a box that is 4 long, 2 wide, and 0.5 tall. Let's do it the base-area way. What is the base, 4 × 2?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- Capsules A and B both have V = _____ because _____ × _____ × _____ = _____ × _____ × _____. I'd choose _____ because _____.

Word bank Volume | Rectangular prism | Cubic units | Dimensions | Base area | 3 | volume | same | different | dimensions | shape | length

WATCH OUT!

A common mistake

A common mistake in Volume of Rectangular Prisms is rounding a fractional edge length to the nearest whole number before multiplying, instead of using the exact fraction or decimal. For a box that is 2 ft × 1.5 ft × 1 ft, a student might round 1.5 down to 1 and compute 2 × 1 × 1 = 2 ft³, but the correct volume uses the exact edge length: 2 × 1.5 × 1 = 3 ft³

LESSON 5.10

Volume of Rectangular Prisms

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS Your class is building a time capsule to bury on school grounds. The capsule is a rectangular prism that is 2 feet long, 1.5 feet wide, and 1 foot tall. You need to figure out how much stuff can fit inside — that means calculating the volume!

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the volume of each rectangular prism.

Length	Width	Height	Volume
8 in	5 in	2 in	
4 cm	4 cm	4 cm	
10 ft	3 ft	6 ft	

HINT Each row lists a length, width, and height. The blank Volume cell needs a number with cubic units.

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Sort these boxes from SMALLEST volume to LARGEST volume.

Groups:

_____ 4 × 4 × 4

_____ 10 × 2 × 3

_____ 6 × 5 × 3

_____ 2 × 2 × 5

_____ 7 × 3 × 4

HINT Each card gives a box's three measurements. Multiply all three to get its volume, then sort by that total.

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

What is the volume of a rectangular prism with l = 4 in, w = 3 in, h = 5 in?

(A) 12 in³
(B) 24 in³
(C) 60 in³
(D) 60 in²

HINT Spot the three edges: l = 4 in, w = 3 in, h = 5 in. Volume fills the whole box, so all three matter.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

A box has a volume of 36 cm³. Its length is 6 cm and width is 3 cm. What is the height?

(A) 2 cm
(B) 3 cm
(C) 6 cm
(D) 12 cm

HINT You're given the volume (36 cm³) plus two edges (6 cm and 3 cm). The height is what's missing.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

How is volume different from area?

(A) Volume uses cubic units; area uses square units

(B) Volume uses square units; area uses cubic units

(C) Volume only applies to cubes; area applies to all shapes

(D) There is no difference

HINT Think dimensions: area covers a flat 2D surface, while volume fills a 3D space.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Dimensions** A box is 5 in long, 4 in wide, and 2 in tall.
2. **Formula** V = length × width × height
3. **Calculate** V = 5 + 4 + 2 = 11 cubic in
4. **Answer** Volume = 11 cubic in

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Check Step 3: should you add the dimensions or multiply them?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Capsules A and B have the same volume (3 ft³) but different dimensions. Why? Which design would you choose and why?

SENTENCE STARTER Capsules A and B both have V = ___ because ___ × ___ × ___ = ___ × ___ × ___. I'd choose ___ because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.10

Volume of Rectangular Prisms

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

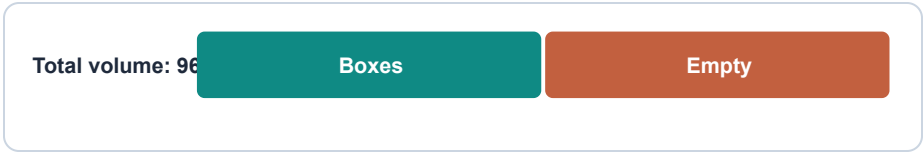
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PART 1 Understand the idea

models and meaning

1 **Use the model to solve. Show your work.**



The container is half full. How many cubic feet of space are left?

SHOW YOUR WORK

Answer: _____

2 **Complete the table. Show how you found each value.**

Find the missing dimension or volume for each container.

Container	Length	Width	Height	Volume
Box A	12 in	5 in	3 in	
Box B	7 cm		4 cm	168 cm³
Box C		8 ft	5 ft	240 ft³

SHOW YOUR WORK

3 **Write the name of the correct group on each line.**

A pool is 12 ft × 8 ft × 5 ft. Sort the statements from FALSE to TRUE.

Groups:

_____ The pool volume is 96 ft³

_____ The pool holds 400 ft³

_____ The pool volume is 480 ft³

_____ V = 12 × 8 × 5

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Two boxes: Box A is 10 × 5 × 4 inches. Box B is 8 × 6 × 5 inches. Which has more volume and by how much?

(A) Box B by 40 in³

(B) Box A by 40 in³

(C) Box B by 60 in³

(D) They're equal

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What is the volume of a prism with edge lengths 1/2 m, 3 m, and 4 m?

(A) 3 m³

(B) 6 m³

(C) 7.5 m³

(D) 12 m³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Dimensions** A box is 6 in long, 4 in wide, and 1/2 in tall.
2. **Formula** V = length × width × height
3. **Calculate** V = 6 × 4 × 1/2, and the student dropped the 1/2: V = 6 × 4 = 24 cubic in
4. **Answer** Volume = 24 cubic in

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Capsules A and B have the same volume (3 ft³) but different dimensions. Why? Which design would you choose and why?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.10

Volume of Rectangular Prisms

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS Your class is building a time capsule to bury on school grounds. The capsule is a rectangular prism that is 2 feet long, 1.5 feet wide, and 1 foot tall. You need to figure out how much stuff can fit inside — that means calculating the volume!

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Two L-shaped containers are each made from two rectangular prisms joined together. Find the total volume of each.

Container	Prism 1 Dimensions	Prism 2 Dimensions	Total Volume
L-shape A	6 × 4 × 3	6 × 2 × 3	
L-shape B	10 × 5 × 2	4 × 5 × 4	

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A fish tank measures 2 1/2 ft by 2 ft by 3 ft. What is its volume?

(A) 7.5 ft³

(B) 12 ft³

(C) 15 ft³

(D) 30 ft³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

3 **Circle the letter of the best answer. Show how you know.**

A prism with a 3/4 ft by 2 ft base has a volume of 9 ft³. What is its height?

(A) 4 ft

(B) 6 ft

(C) 9 ft

(D) 12 ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

PART 3 Explain and correct

reasoning

4 Find the mistake. Explain it, then show the correct work.

Find the Volume Error

1. **Dimensions** l = 5 m, w = 3 m, h = 2.5 m
2. **Formula** V = l × w × h
3. **Calculate** V = 5 × 3 × 2.5 = 5 × 5.5 = 27.5 m³
4. **Answer** Volume = 27.5 m³

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

5 Answer in complete sentences.

A toy company ships action figures in boxes that are 4 × 3 × 6 inches. A shipping crate is 24 × 12 × 18 inches. How many action figure boxes fit in the crate? Explain your reasoning step by step.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Capsules A and B have the same volume (3 ft³) but different dimensions. Why? Which design would you choose and why?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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